

DAY-2 MACHINE TEST – LINUX FILESYSTEM & NAVIGATION PART A – MULTIPLE CHOICE QUESTIONS

1. What is a Linux Filesystem?

- A. Database
- B. Structure used to organize files and directories
- C. Programming Language
- D. Web Browser

2. Which symbol represents the Linux Root Directory?

- A. ~
- B. ..
- C. /
- D. .

3. Which directory usually stores user files?

- A. /etc
- B. /home
- C. /bin
- D. /tmp

4. Which directory contains configuration files?

- A. /var
- B. /usr
- C. /etc
- D. /home

5. What does the clear command do?

- A. Delete files
- B. Clear terminal screen
- C. Create directory
- D. Exit terminal

6. What is the purpose of sudo?

- A. Change directory
- B. Execute command with administrator privileges
- C. Show user
- D. Show hostname

7. What does pwd stand for?

- A. Print Working Data
- B. Present Working Directory
- C. Personal Work Directory
- D. Path Working Directory

8. Which command lists files and folders?

- A. cd
- B. pwd
- C. **ls**
- D. rm

9. Which command shows hidden files?

- A. ls -l
- B. **ls -a**
- C. ls -R
- D. pwd

10. Which command shows detailed file information?

- A. **ls -l**
- B. ls -a
- C. ls
- D. pwd

11. What does ls -la display?

- A. Files only
- B. Hidden files only
- C. **Detailed list including hidden files**
- D. Directories only

12. Which command changes directory?

- A. ls
- B. pwd
- C. **cd**
- D. mkdir

13. What does cd .. do?

- A. Go home
- B. Go to root
- C. **Move to parent directory**
- D. Delete directory

14. What does cd ~ do?

- A. Go root
- B. Go parent directory
- C. **Go home directory**
- D. Exit terminal

15. Which path is an absolute path?

- A. Documents
- B. Desktop
- C. Downloads
- D. **/home/ubuntu/Documents**

16. Which command creates a directory?

- A. touch
- B. **mkdir**
- C. rm
- D. pwd

17. Which option creates parent directories automatically?

- A. mkdir -r
- B. mkdir -f
- C. **mkdir -p**
- D. mkdir -a

18. What does ls -R do?

- A. Remove files
- B. Show hidden files
- C. **Recursive listing**
- D. Rename files

19. Which command removes an empty directory?

- A. rm
- B. **rmdir**
- C. pwd
- D. ls

20. Which command removes a directory and its contents? A. **rm -r**

- B. rmdir
- C. mkdir
- D. ls

PART B – THEORY QUESTIONS

1. Define Linux Filesystem.

The Linux Filesystem is the organized structure that Linux uses to store, manage, and retrieve files on a storage drive.

2. Explain the Root Directory (/).

The Root Directory, represented by a single forward slash (/), is the very starting point of the entire Linux filesystem.

3. What is the purpose of /home?

The /home folder is the place where individual users store their personal files. Inside /home, every user gets their own dedicated folder to keep their documents, downloads, desktop items, and personal settings safe from other users.

4. What is the purpose of /etc?

It serves as the central storage space for all system-wide configuration files

5. Explain the clear command.

The clear command wipes the visible text off your terminal screen.

6. What is sudo and why is it used?

It is a command prefix used to run programs with administrative privileges

7. Explain the pwd command.

The pwd command stands for "Print Working Directory." When you type it into the terminal, it outputs the exact, full path of the folder you are currently working inside

8. Differentiate between ls and ls -la.

- ls is the standard command that lists only the basic, visible files and folders inside your current directory.
- ls -la tells the system to use a long listing format (showing extra details like file size, permissions, and creation dates) and to show all files, including hidden system files.

9. Explain cd .., cd ~ and cd /.

- cd .. moves you one level backward into the parent folder (the directory directly above your current one).
- cd ~ instantly takes you back to your personal home folder, no matter where you currently are in the system.
- cd / takes you directly to the very bottom of the system—the Root Directory.

10. What is an Absolute Path?

An Absolute Path is a complete, explicit address that points to a file or folder starting from the very root (/) of the system.

11. What is a Relative Path?

A Relative Path is a directional path that points to a file or folder based on where you are currently standing right now. It does not start with a /.

12. Explain the purpose of mkdir -p.

The mkdir -p command allows you to create a pathway of nested folders all at once. The -p flag stands for "parents "

13. Explain the purpose of ls -R.

The ls -R command performs a Recursive listing. It doesn't just show the files in your current folder; it forces the terminal to drill down and show you all the files inside every single sub-folder and sub-sub-folder under your current location.

14. Differentiate between rmdir and rm -r.

- rmdir is a cautious command that can *only* delete a folder if that folder is completely empty. If there is even one file inside, it will refuse to run.
- rm -r stands for "remove recursive." It forcefully deletes a folder along with every single file and sub-folder trapped inside it.

15. Why is rm -rf considered dangerous?

The rm -rf command tells the system to recursively delete a directory and force the deletion without asking for confirmation or warnings. If a user accidentally types this at the Root Directory (sudo rm -rf /), Linux will silently and permanently erase the entire operating system, files, and connected drives with no way to hit "undo."

PART C – PRACTICAL EXERCISES

Exercise 1 – Navigation Commands

pwd

ls

ls -la

cd ~

pwd

cd /

pwd

Take screenshots of all outputs.

Exercise 2 – Directory Creation

mkdir Linux

mkdir Docker

mkdir AWS

mkdir Kubernetes

Verify using ls.

Exercise 3 – Nested Directory Creation

mkdir -p devops/day2/practice

Verify using:

ls -R devops

Exercise 4 – Navigation Practice

cd devops

pwd

cd day2

pwd

cd practice

pwd

cd ../..

Exercise 5 – Directory Removal

mkdir test

mkdir demo

rmdir test

rm -r demo

Verify using ls

MINI INTERVIEW / VIVA QUESTIONS

1. What is Linux Filesystem?
2. What is the Root Directory?
3. Difference between ls and ls -la?
4. What does cd .. do?
5. What does cd ~ do?
6. What is an Absolute Path?
7. What is a Relative Path?
8. What does mkdir -p do?
9. Difference between rmdir and rm -r?
10. Why should DevOps engineers be careful with rm -rf?